$x \ll y \iff \exists n \in \mathbb{N}^* / y = x^n$.

Ref: for $n = 1 \Rightarrow x = x^1$

$\Rightarrow x \ll x$.

Transitive: let $x, y, z \in \mathbb{N}^*$

$x \ll y \wedge y \ll z \Rightarrow$

$\exists n / y = x^n \wedge \exists m \; z = y^m$

$\Rightarrow z = (x^n)^m = x^{n \cdot m}$

$\Rightarrow \exists k = n \cdot m \; / \; z = x^k$

$\Rightarrow x \ll z$

Anti-sym:

$(x \ll y \wedge y \ll x)$

$\Rightarrow y = x^n \wedge x = y^m \quad n \in \mathbb{N}^*$

$\Rightarrow x = y^m = (x^n)^m = x^{n \cdot m}$ $m \in \mathbb{N}^*$

$\Rightarrow n = m = 1$.

$\Rightarrow n \cdot m = 1 \Rightarrow n = 1 \wedge m = 1$

$\Rightarrow y = x \Rightarrow \ll$ ~~is a~~

$\ll$ is a partial order.

$x = 3 \wedge y = 5$

$? n \in \mathbb{N}^* \quad 5 = 3^n ?$

impossible.

$\Rightarrow \ll$ is partial order.